



Toyota Motor Sales, U.S.A., Inc.
19001 South Western Avenue
P.O. Box 2991
Torrance, CA 90509-2991
(310)618-4000
(310)618-7800 Fax

TO: ALL TOYOTA DEALER PRINCIPALS,
SERVICE MANAGERS, PARTS MANAGERS

SUBJECT: SPECIAL SERVICE CAMPAIGN – P01 (A/C EXPANSION VALVE)

Under certain conditions, particularly in high temperature and high humidity areas, the air conditioning expansion valve in certain Toyota models may malfunction. As a result, there will be a gradual reduction of air conditioning efficiency and eventual loss of cooling capability.

Toyota will initiate a Special Service Campaign to inspect the air conditioning system, inject an anti-corrosion lubricant (Toyota ND6 AC Refrigerant Oil Charge) or, if necessary, replace the air conditioning expansion valve on certain 1989 – 1993 Toyota models.

PLEASE BE AWARE THAT RECHARGING OR REPAIRS TO ANY A/C SYSTEM COMPONENT OTHER THAN THOSE DETAILED IN THE TECHNICAL INSTRUCTIONS MUST BE HANDLED AS FOLLOWS:

- * VEHICLE WARRANTY STILL IN EFFECT: SUBMIT REGULAR WARRANTY CLAIM.
- * VEHICLE WARRANTY NO LONGER IN EFFECT: CUSTOMER MUST PAY.

The following information is provided to inform you and your staff of the campaign schedule and your degree of involvement required:

1. Dealer, Owner Notification Letter Mailing Dates

- Dealer and Owner notifications will be mailed according to the following schedule:

To Southeast Toyota Distributor and Gulf States Toyota Distributor Dealerships and Owners: May, 1993

To San Francisco, Los Angeles, Denver, Central Atlantic Toyota Distributor, New York and Kansas City Toyota Regions Dealerships and Owners: June, 1993.

To Portland, Cincinnati, Chicago and Boston Toyota Regions Dealerships and Owners: July, 1993.

NOTE: You will be sent SSC notifications with your dealer's name and address for vehicles which, according to our record, you have in stock. If any vehicles in your stock involved in this program have been sold to a customer prior to receipt of this notification, please contact this known customer and perform the necessary repair.

2. **Identification of Involved Vehicles**

Following are complete listings of qualified vehicles:

MODEL	MODEL YEAR	VDS	INVOLVED VIN RANGE
Camry	1992 – 1993	SK11E	0001001 – 0103996
			U000001 – U164297
		SK12E	0001001 – 0103996
			U000001 – U164297
		SK13E	0001001 – 0103996
			U000001 – U164297
		VK11E	0001001 – 0129016
			U000001 – U048888
		VK12E	0001001 – 0129016
			U000001 – U048888
		VK13E	0001001 – 0129016
			U000001 – U048888
		VK14E	0001001 – 0129016
			U000001 – U048888
		SK11W	U000001 – U164297
		SK12W	U000001 – U164297
		VK12W	U000001 – U048888

2. **Identification of Involved Vehicles** (cont'd)

MODEL	MODEL YEAR	VDS	INVOLVED VIN RANGE
Celica	1990 – 1993	AT86F	0000001 – 0100665
		ST87F	0000001 – 0134938
		ST87N	0000001 – 0134938
		ST87K	0000001 – 0134938
		ST85N	0000001 – 0134938
		ST88P	0000001 – 0023255
4Runner	1990 – 1993	RN27V	0000001 – 0001527
		RN37J	0000001 – 0011233
		RN37W	0000001 – 0011233
		VN29V	0000001 – 0014106
		VN39J	0000001 – 0096750
			8000001 – 8052248
		VN39W	0000001 – 0096750
			8000001 – 8052248
Land Cruiser	1991 – 1992	FJ80W	0000001 – 0051986
	1993	DJ81W	0000001 – 0012752
MR2	1991 – 1993	SW21M	0000001 – 0019446
		SW21N	0000001 – 0019446
		SW22M	0000001 – 0076191
		SW22N	0000001 – 0076191
Previa	1991 – 1992	AC11R	0000001 – 0074543
			1001001 – 1058504
		AC12R	0000001 – 0074543
			1001001 – 1058504
		AC21S	0000001 – 0013972
			1001001 – 1012920
		AC22S	0000001 – 0013972
			1001001 – 1012920

2. **Identification of Involved Vehicles** (cont'd)

MODEL	MODEL YEAR	VDS	INVOLVED VIN RANGE
Supra	1989 – 1992	MA70L	0104941 – 0162266
		MA70J	0104941 – 0162266
		MA71L	0104941 – 0162266
		MA71J	0104941 – 0162266
		MA70M	0104941 – 0162266
		MA70N	0104941 – 0162266
		MA71M	0104941 – 0162266
		MA71N	0104941 – 0162266

3. **Repair Procedures**

- Please refer to the attached Technical Instructions for the applicable vehicle model.

4. **Parts Ordering**

- Necessary parts for this Special Service Campaign must be ordered from your dealer's facing Parts Distribution Center. The anti-corrosion lubricant comes in packages of 12 cans per package only. Many of the vehicles will need only the injection of the anti-corrosion lubricant.
Please order the refrigerant oil cans in multiples of twelve.
- **A SUMMARY LISTING OF THE INVOLVED VEHICLES SOLD BY YOUR DEALERSHIP IS ENCLOSED. THIS LISTING MUST BE UTILIZED TO CALCULATE REPLACEMENT PARTS REQUIREMENTS BY MODEL. YOUR INITIAL PARTS ORDERS MUST NOT BE MORE THAN 5% OF YOUR INVOLVED VEHICLES. SUBSEQUENT ORDERS MUST BE CALCULATED TO MEET ACTUAL WEEKLY DEMAND REQUIREMENTS IN ORDER TO ENSURE CONTINUED GOOD AVAILABILITY OF PARTS.**
- Please refer to the attached Parts Order Section for specific parts information by vehicle model.

5. **Tools and Equipment**

- Please refer to the attached Preparation and Precaution Section to determine if you have the tools and equipment necessary to perform all aspects of the service.

NOTE: One Special Service Tool Part No. TOY219721 (AC Oil Injection Kit) will be shipped to each dealer during May, 1993 free of charge. Additional tool kits can be ordered through Owantana Tool Company for \$28.91 each.

6. **Reimbursement Procedures**

- Submit Special Service Campaign claims following the procedures described in the Toyota Warranty Policy & Procedures Manual.

The operational codes to be used for this campaign are listed below.

Op. Code	Description	Flat Rate Hour
350041	Add Toyota ND6 AC Refrigerant Oil Charge	0.6 hr/veh
350042	Add Toyota ND6 AC Refrigerant Oil Charge after recovery and recharge of refrigerant	1.0 hr/veh
350043	Add Toyota ND6 AC Refrigerant Oil Charge after confirming no leaks at drain hose	0.9 hr/veh
350044	Replace one expansion valve for all models	2.2 hrs/veh
350045	Replace both front & rear expansion valves for Previa	2.6 hrs/veh
350046	Customer reimbursement on previous repairs	Actual Expense

Note: The above operation codes include 0.1 flat rate hour for administrative expenses.

7. **Customer Reimbursements**

Please reimburse any owners who have previously paid to replace the air conditioning expansion valve.

- Enter the reimbursement charges as a CR sublet type.
- Use the customer's invoice number as the sublet invoice number.
- Enter the reimbursement amount in the sublet amount column.
- Provide a brief description of the reason for the reimbursement.
- Use the customer reimbursement operation code number 350046.

Please review this entire Special Service Campaign package with your Service and Parts staff to familiarize them with the proper step by step procedure and implement this Special Service Campaign to ensure maximum customer satisfaction.

Your cooperation in this matter would be greatly appreciated.

TOYOTA MOTOR SALES, U.S.A., INC.

Dear Toyota Owner:

Toyota would like to announce a special program for certain vehicles equipped with Toyota air conditioning.

WHAT IS THE CONDITION?

Under certain high temperature and high humidity conditions, the air conditioning expansion valve may malfunction. This malfunction will be evidenced by a gradual reduction of air conditioning efficiency and eventually a loss of cooling capability.

WHAT WILL TOYOTA DO?

If your vehicle is equipped with Toyota air conditioning, Toyota will inspect and, if necessary, repair your vehicle's air conditioning system as follows:

- If the system is functioning correctly, a special anti-corrosion lubricant will be injected into the system to protect the components at NO COST to you.
- If the expansion valve is not performing correctly, it will be replaced and the system recharged at NO COST to you.
- If your vehicle's air conditioning system warranty is still in effect, any necessary recharging or other A/C system repairs will be performed at NO COST to you within the guidelines of Toyota's vehicle warranty policy.
- If recharging or other repairs are necessary and your vehicle's air conditioning system warranty has expired, there may be some items for which you will be charged. Your Toyota dealer will provide you with an estimate before repairs are performed.

WHAT SHOULD YOU DO?

Make an appointment with your Toyota dealer at your earliest convenience to have this inspection performed. Please do not delay as this inspection should be performed as soon as possible. Present this notice to your Toyota dealer when you bring your vehicle in for the inspection.

Depending on the nature of the repair, the work will take from 30 minutes to 3 hours. However, it may be necessary to make your vehicle available for a longer period of time, depending upon the dealer's work schedule.

If you have previously paid for the replacement of your expansion valve, please present the invoice to your dealer for reimbursement.

WHAT IF YOU HAVE OTHER QUESTIONS?

Please contact your local Toyota dealer or:

Toyota Customer Assistance Center
1 (800) 331-4331

We would like to thank you for your cooperation and wish to apologize for any inconvenience this may cause you.

Thank you for driving a Toyota.

Sincerely,

TOYOTA MOTOR SALES, U.S.A., INC.
CORPORATE SERVICE DEPARTMENT

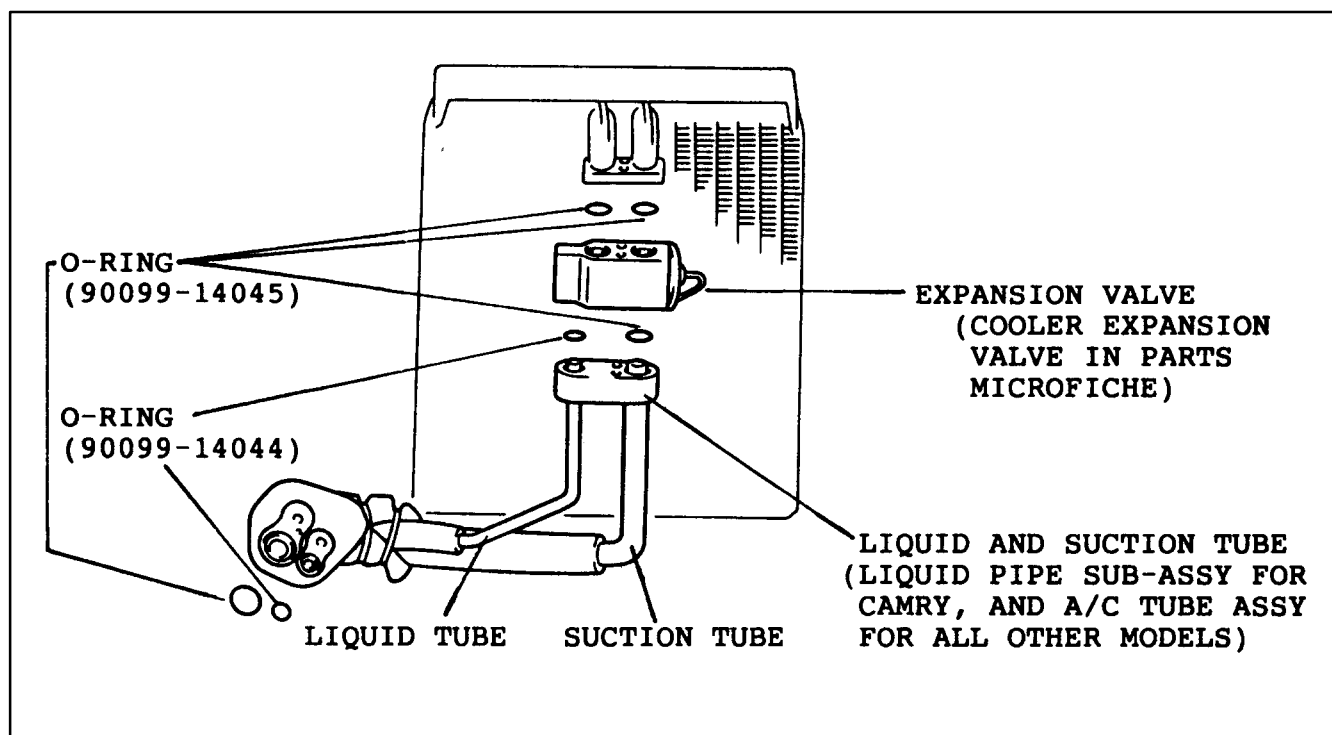
PARTS ORDER INFORMATION

- Necessary parts for this Special Service Campaign must be ordered from your dealer's facing Parts Distribution Center. The anti-corrosion lubricant comes in packages of 12 cans per package only. Many of the vehicles will need only the injection of the anti-corrosion lubricant. **Please order the refrigerant oil cans in multiples of twelve.**
- Please utilize the attached summary listing of the involved vehicles sold by your dealership to calculate replacement parts requirements by model. Your initial parts orders must not be more than 5% of your involved vehicles. Subsequent orders must be calculated to meet actual weekly demand requirements in order to ensure continued good availability of parts.

Toyota ND6 AC Refrigerant Oil Charge must be used during this Special Service Campaign.

Model	Part Name	Part Number	Qty/Vehicle
All	Toyota ND6 AC Refrigerant Oil Charge	00263-09106	One Can

Expansion Valve, Liquid and Suction tube, O-rings.



The number of O-rings to be replaced depends on whether the liquid and suction tube is replaced. O-rings between the expansion valve and liquid and suction tube are supplied as part of the liquid and suction tube. For details, see the parts list of specific vehicles on the following tables.

PARTS ORDER INFORMATION (cont'd)

Expansion Valve, Liquid and Suction tube, O-rings. See Appendix for List of VIN involvement for tube replacement.

Model	Part Name	Part Number	Qty/Vehicle	
			With Tube Replacement	Without Tube Replacement
Camry	Cooler Expansion Valve	88515-33010	1	1
	Liquid Pipe Sub-Assy	88706-33020	1	0
	O-ring	90099-14045	3	4
	O-ring	90099-14044	1	2
Celica	Cooler Expansion Valve	88515-33010	1	1
	A/C Tube Assy	88710-20200	1	0
	O-ring	90099-14045	3	4
	O-ring	90099-14044	1	2
4Runner	Cooler Expansion Valve	88515-33010	1	1
	(1) A/C Tube Assy	88710-89122	1	0
	O-ring	90099-14045	3	4
	O-ring	90099-14044	1	2
	(2) O-ring	90099-14056	1	0
Land Cruiser	Cooler Expansion Valve	88515-33010	1	1
	A/C Tube Assy	88710-60230	1	0
	O-ring	90099-14045	3	4
	O-ring	90099-14044	1	2
MR2	Cooler Expansion Valve	88515-33010	1	1
	(3) A/C Tube Assy	88710-17070	1	0
	O-ring	90099-14045	3	4
	O-ring	90099-14044	1	2

PARTS ORDER INFORMATION (cont'd)

Expansion Valve, Liquid and Suction tube, O-rings. See Appendix for List of VIN involvement for tube replacement.

Model	Part Name	Part Number	Qty/Vehicle	
			With Tube Replacement	Without Tube Replacement
Previa (4)	FR	Cooler Expansion Valve	*	1
		O-ring	*	3
	AC	O-ring	*	1
	RR	Cooler Expansion Valve	*	1
		O-ring	*	3
	AC	O-ring	*	2
		O-ring	*	1
Supra (5)	Cooler Expansion Valve		*	1
	O-ring		*	4
	O-ring		*	2

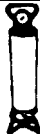
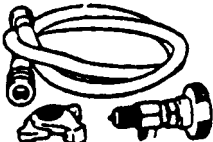
- (1) Pressure Switch is **NOT** included in A/C Tube Assy (88710-89122) and original Pressure Switch must be reused.
- (2) This is the O-ring (90099-14056) between A/C Tube and Pressure Switch.
- (3) Replace the Pressure Switch with A/C Tube Assy. Pressure Switch is included in A/C Tube Assy. (88710-17070)
- (4) Replacement of Liquid and Suction Tube is **NOT** required on Previa.
- (5) Replacement of Liquid and Suction Tube is **NOT** required on Supra.

* Not Required.

PREPARATION AND PRECAUTION

1. REQUIRED TOOLS

Tools:

Air Conditioning Service Tool Set		Refrigerant Recovery Machine	
Vacuum Pump Assembly		Charging Cylinder	
Gas Leak Detector Assembly		AC Oil Injector Kit (TOY219721)	
Socket Hexagon Wrench 5 mm		Torque Wrench	
		Standard Tools	

STOP VALVES MUST BE USED TO MEET YOUR LOCAL STATE REGULATIONS

Other Necessities: * Dry bulb thermometers

* Repair Manuals for applicable models and model years

Model	Pub. No.	Model	Pub No.
Camry	1992 Model: RM222U2 1993 Model: RM307U2	MR2	1991 Model: RM179U2 1992 Model: RM252U2 1993 Model: RM285U2 (To Jul, 1992) 1993 Model: RM308U2 (From Aug, 1992)
Celica	1990 Model: RM149U 1991 Model: RM197U2 1992 Model: RM251U2 1993 Model: RM305U2	Previa	1991 Model: RM178U 1992 Model: RM255U
4Runner	1990 Model: RM143U2 1991 Model: RM202U2 1992 Model: RM257U2 1993 Model: RM309U2	Supra	1989 Model: RM115U 1990 Model: RM150U 1991 Model: RM198U 1992 Model: RM253U
Land Cruiser	1991 Model: RM180U 1992 Model: RM258U 1993 Model: RM301U		

2. PRECAUTIONS

(a) Handling precautions for refrigerant.

- Do not handle refrigerant in an enclosed area or near an open flame.
- Always wear protective glasses when working with or around refrigerant or refrigeration equipment.
- Take care that the liquid refrigerant does not get in your eyes or on your skin.
- If liquid refrigerant gets in your eyes or on your skin:
 - i. Wash the area with plenty of cool water.
Caution: Do not rub your eyes or skin.
 - ii. Apply clean petroleum jelly to the skin.
 - iii. Go immediately to a physician or hospital for professional treatment.
Caution: Do not attempt to treat yourself.

(b) Handling precautions for refrigerant oil charge can.

- Before starting the operation, please read and observe all cautions and warnings written on the container.
- Never heat the container or expose it to open flame.
- Take care not to drop the container and not to apply physical shock.

(c) Technician **MUST BE** either MACS or ASE certified.

(d) Obey your local state regulations when applicable.

3. PREPARATION

Before starting work, place protective covers on the:

- * fender
- * seats
- * floor mats

4. RECOVERY OF REFRIGERANT IN REFRIGERANT SYSTEM

When replacing parts or if the refrigerant quantity is excessive, recover the refrigerant first. Always use a recovery machine when recovering the refrigerant.

- NOTE:**
- Whenever using the recovery machine, always follow the directions given in the instruction manual.
 - When the refrigerant is recovered, the compressor oil is also being recovered with the refrigerant. When recharging a system that has been evacuated, **DO NOT** reuse the old compressor oil, measure the amount of compressor oil removed and install the same amount of **NEW Toyota ND OIL 6 Compressor Oil** into the system.

4. RECOVERY OF REFRIGERANT IN REFRIGERANT SYSTEM (cont'd)

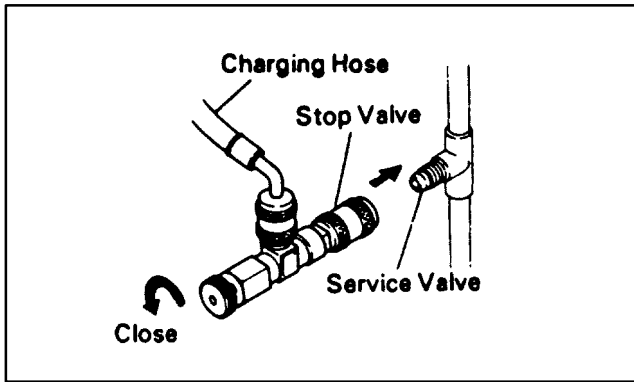


Figure 1

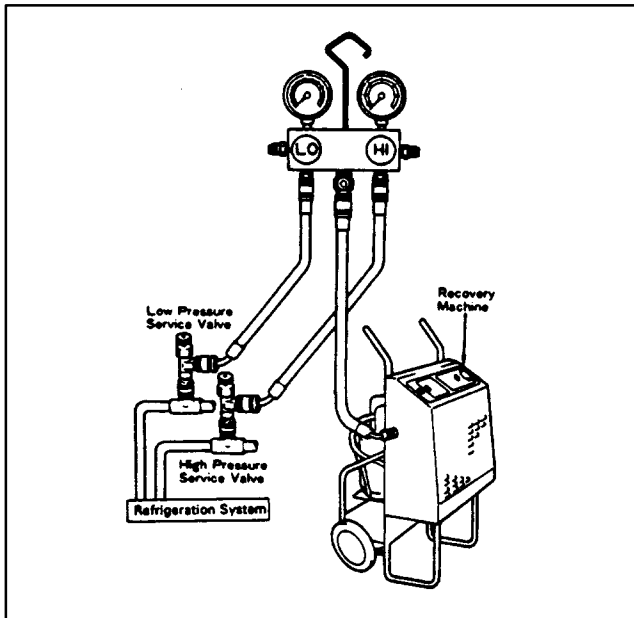


Figure 2

- (1) Installation of manifold gauge set to service valves.

NOTE: To prevent releasing refrigerant, use charging hoses with a stop valve when installing the manifold gauge set to service valves on the refrigerant line.

(Figure 1)

- (i) Connect charging hoses with a stop valve to manifold gauge set. Tighten the nuts by hand.

CAUTION: – Do not connect the wrong hoses to the high pressure and the low pressure sides. Please refer to the charging hoses locations shown under applicable model on Page P&P-5.

- To prevent loosening the nuts, do not apply compressor oil to seat of the connection.

- (ii) Close hand valves of both stop valves.
- (iii) Close both hand valves of manifold gauge set.
- (iv) Remove caps from service valves on refrigerant line.
- (v) Connect stop valves to service valves. Tighten the nuts by hand.

CAUTION: – Do not connect the wrong hoses to the high pressure and the low pressure sides.
– To prevent loosening the nuts, do not apply compressor oil to the seat of the connection.

- (vi) Open hand valves of both stop valves.
- (2) Recovering refrigerant from refrigeration system.
 - (i) Connect the center hose to recovery machine.
 - (ii) Operate the recovery machine.
 - (iii) Open both high and low hand valves of manifold gauge set.

(Figure 2)

NOTE: Use the hose with self-sealing stop valve to prevent air from getting into the center hose while performing the operation.

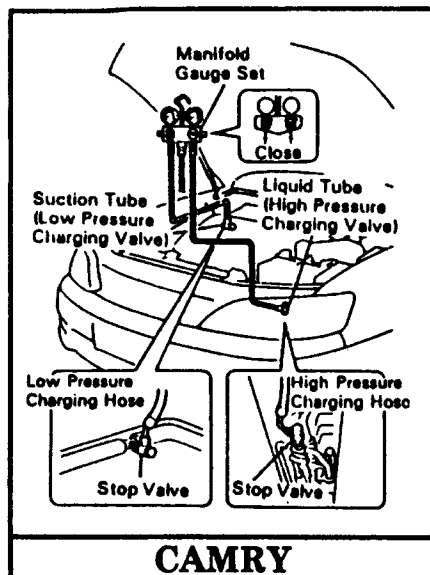
- (3) Stop recovery machine when recovering has finished.

4. RECOVERY OF REFRIGERANT IN REFRIGERANT SYSTEM (cont'd)

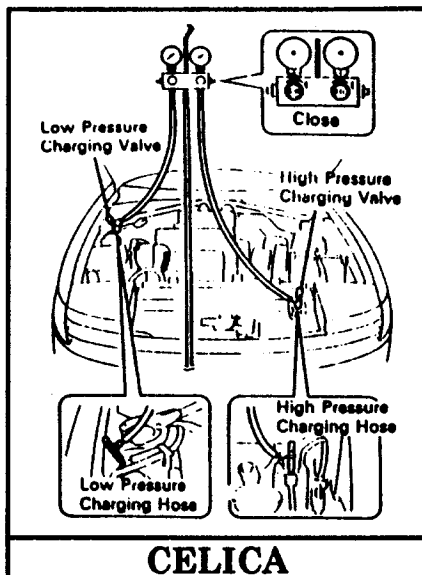
- (4) Removing the manifold gauge set from service valves.
 - (i) Close both hand valves of manifold gauge set.
 - (ii) Close hand valves of both stop valves.
 - (iii) Disconnect stop valves from service valves on refrigerant line.
 - (iv) Install caps to service valves on refrigerant line.

5. REFRIGERANT CHARGING VOLUME

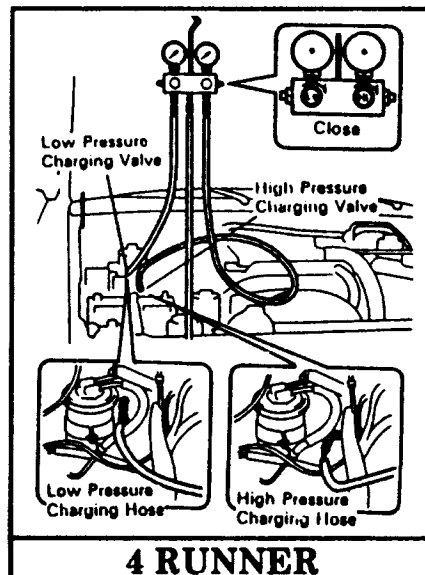
Refer to each vehicle model section for the correct charging volume specification.



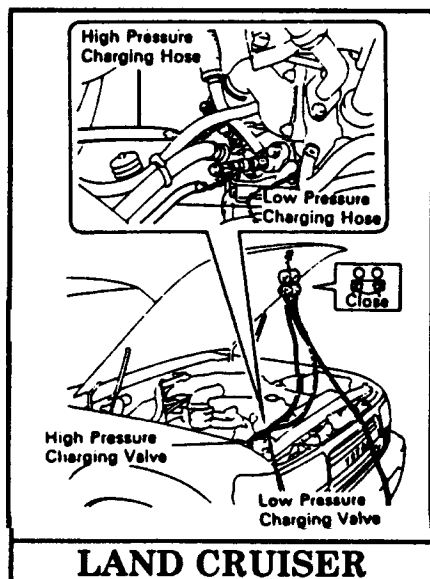
CAMRY



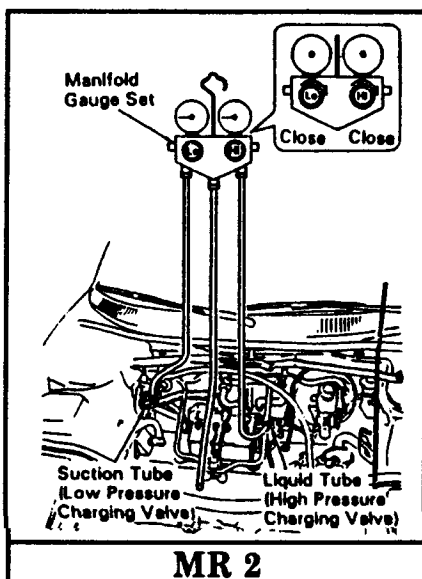
CELICA



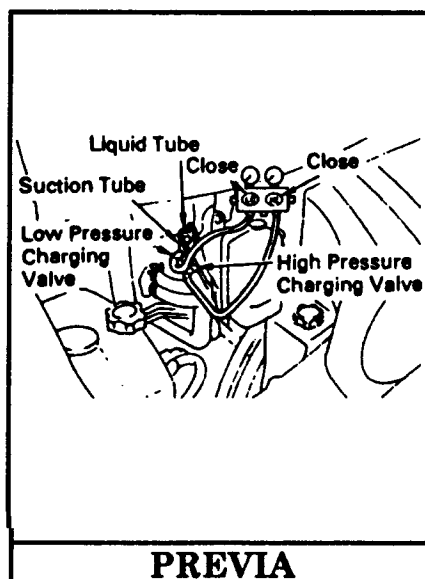
4 RUNNER



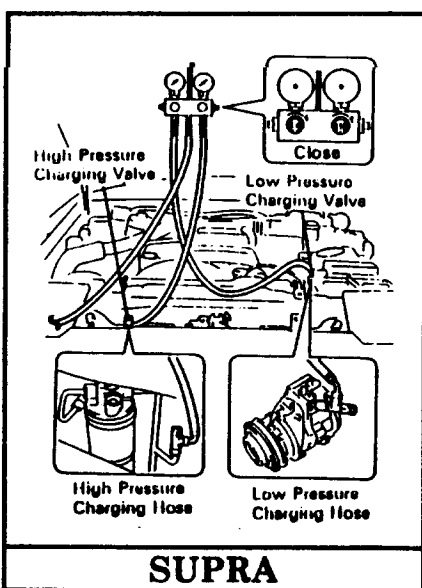
LAND CRUISER



MR 2



PREVIA



SUPRA

APPENDIX

Affected vehicles requiring inspection and/or replacement of liquid and suction tube.

NOTE: When replacing the expansion valve, if the vehicle falls within following VIN ranges, also replace the liquid and suction tube provided that you have visually confirmed that the spacers at the end of the liquid and suction tube are zinc plated steel (gold color).

MODEL	MODEL PREFIX	INVOLVED VIN RANGE
CAMRY 1992 – 1993	JT2SK11E*	*0001001 – *0089378
	4T1SK11E*	*U000001 – *U131913
	JT2SK12E*	*0001001 – *0089378
	4T1SK12E*	*U000001 – *U131913
	JT2SK13E*	*0001001 – *0089378
	4T1SK13E*	*U000001 – *U131913
	JT2VK11E*	*0001001 – *0108289
	4T1VK11E*	*U000001 – *U036966
	JT2VK12E*	*0001001 – *0108289
	4T1VK12E*	*U000001 – *U036966
	JT2VK13E*	*0001001 – *0108289
	4T1VK13E*	*U000001 – *U036966
	JT2VK14E*	*0001001 – *0108289
	4T1VK14E*	U000001 – *U036966
	4T1SK11W*	*U000001 – *U131913
	4T1SK12W*	*U000001 – *U131913
	4T1VK12W*	*U000001 – *U036966
Celica 1990 – 1993	JT2AT86F*	*0000001 – *0083990
	JT2ST87F*	*0000001 – *0112501
	JT2ST87N*	*0000001 – *0112501
	JT5ST87K*	*0000001 – *0112501
	JT2ST85N*	*0000001 – *0112501
	JT2ST88P*	*0000001 – *0020394

APPENDIX (cont'd)

MODEL	MODEL PREFIX	INVOLVED VIN RANGE
4Runner 1990 – 1993	JT3RN27V*	*0000001 – *0001527
	JT3RN37J*	*0000001 – *0010575
	JT3RN37W*	*0000001 – *0010575
	JT3VN29V*	*0000001 – *0011908
	JT3VN39J*	*0000001 – *0083650
	JT3VN39J*	*8000001 – *8048908
	JT3VN339J*	*8000001 – *8048908
	JT3VN39W*	*0000001 – *0083650
	JT3VN39W*	*8000001 – *8048908
Land Cruiser 1991 – 1993	JT3FJ80W*	*0000001 – *0048813
	JT3DJ81W*	*0000001 – *0005436
MR2 1991 – 1993	JT2SW21M*	*0000001 – *0018913
	JT2SW21N*	*0000001 – *0018913
	JT2SW22M*	*0000001 – *0071937
	JT2SW22N*	*0000001 – *0071937